# Arthur Borem

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## SKILLS

**Tech:** TypeScript/JavaScript, Python, Scala; React, Node.js, Django, GraphQL; REST APIs; PostgreSQL, DynamoDB, Redis.
**Specialization:** privacy-preserving systems, web tracking, data subject rights compliance and tooling, OAuth, data access/portability.

## EDUCATION

**PhD in Computer Science** (ongoing)**.** University of Chicago. *Chicago, IL.* **Sep 2021 – Jun 2026 (exp)**
Dissertation: Tooling for GDPR/CCPA data subject rights (access, portability), privacy-preserving data handling.
Awards: John Crerar Fellowship, NSF Graduate Research Fellowship Program (Honorable Mention).

**MS in Computer Science.** University of Chicago. *Chicago, IL.* **Sep 2021 – Jun 2024**

**BS in Computer Science.** Brown University. *Providence, RI.* **Aug 2016 – May 2020**

## EXPERIENCE

**Data Transfer Initiative**. Software Engineer Co-op. *Remote.* **Jul 2025 – Sep 2025**

- Led development and launch of Pardner (github.com/dtinit/pardner), Python package, enabling user-authorized (OAuth 2.0) data portability with standardized data formats between companies.
- Built automated standardization and validation of personal data category formats from different services, eliminating manual field-mapping for data scientists and engineers working on portability.
- Deployed Django-based data donation web-app, reducing participant donation time from multi-day communications with researchers to automated data transfers that last milliseconds.
- Defined data flows and documentation for partner organizations to run privacy reviews.

**Asana**. Software Engineer. *San Francisco, CA.* **Jun 2020 – Sep 2021**

- Launched video recording feature (React, GraphQL, AWS) to millions of users in collaboration with Vimeo engineers; 80%+ enterprise adoption and earned Fast Company joint-venture award.
- Launched text-based video search by indexing transcripts in Elasticsearch.
- Built backend (Typescript, Scala) for first AI-powered recommendation system for messages, delivering context-aware suggestions for collaborators, projects, and tasks directly in the text composer UI.
- Created a roadmap and assigned tasks to three engineers for the implementation of “Do not Disturb” notifications; partnered with product, finance, and design to run experiments to minimize churn risk.
- Maintained and optimized email infrastructure, delivering and ingesting 5k+/day emails.

**Lyft**. Software Engineer Intern. *San Francisco, CA.* **Jun 2019 – Aug 2019**

- Reduced enterprise support tickets by deploying a Lyft Business system health REST API (Python, Redis, DynamoDB) by monitoring and aggregating response time latency percentiles and endpoint error rates.
- Iteratively designed API schema via direct collaboration with client support engineers and customers.

## PRIVACY TOOLS & RESEARCH

**Data Subject Rights Exploration Platform**. University of Chicago. **2025**

- Deployed privacy-preserving data processing pipeline entirely in a user's browser without increasing latency by parallelizing (via web workers) parsing and redaction tasks and using IndexedDB.
- Designed data visualization features (e.g., mapped location tracking) to provoke users to reflect on privacy.

**Repository of Data Access Request Exports.** University of Chicago. **2025**

- Catalogued DSAR data from 200+ companies, measuring response latency and personal data labeling.
- Generated schemas for exports of 60+ platforms: github.com/UChicagoSUPERgroup/dsar-export-schemas

**Ad-Blocker Browser Breakage Detection**. University of Chicago. **2024**

- Published taxonomy of ad-blocker-induced website breakage (e.g., broken login) by scraping and analyzing thousands of user complaints/reviews; as recognized by Brave Browser and uBlockOrigin.
- Mapped how anti-tracking and ad-blocking tools interact with filter-list rules and cause breakage.

**Personal Data Annotation Tool**. University of Chicago. **2023**

- Led team of 4 to build privacy-preserving React-based web-app for users to interact with, visualize, and annotate their personal data obtained from making data subject access requests (DSARs).
- Eliminated multi-second freezes by paginating large JSON files with thousands of complex nested objects.
- Developed award-winning policy recommendations by analyzing 500+ annotations and DSAR files.

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## TALKS

How Consumers React When Exercising Their Right of Access **2026**
***French Data Protection Authority (CNIL, Research@LINC)***

Enabling Data Portability in a Research Setting **2025**
***DjangoCon US***.

Designing a Data Subject Access Rights Tool **2024**
*USENIX Privacy Engineering Practice and Respect (**PEPR**).*

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## PUBLICATIONS

**Arthur Borem**, Alice Lin, Olufunmilola Obielodan, Michelle Mazurek, Blase Ur. " Unpacking the Data Subject Access Request: Analysis of Request Mechanisms and Personal Data Files Across 250 Platforms." *In submission, 2026*.

Kevin Bryson, **Arthur Borem**, Phoebe Moh, Omer Akgul, Michelle L. Mazurek, Blase Ur. "Evaluating and Taxonomizing Ad Transparency Systems." *IEEE Symposium on Security and Privacy (**S&P 2025**).*

**Arthur Borem**, Elleen Pan, Olufunmilola Obielodan, Aurelie Roubinowitz, Luca Dovichi, Michelle L. Mazurek, Blase Ur. "Data Subjects' Reactions to Exercising Their Right of Access." *USENIX Security Symposium (**USENIX Security 2024**).* **Privacy Papers for Policy Makers Student Award.**

Galen Harrison, Kevin Bryson, Ahmed E. B. Bamba, Luca Dovichi, Aleksander H. Binion, **Arthur Borem**, Blase Ur. "JupyterLab in Retrograde: Contextual Notifications That Highlight Fairness and Bias Issues for Data Scientists." *ACM Conference on Human Factors in Computing Systems (**CHI 2024**).* **Best Paper Award.**

Alexandra Nisenoff, **Arthur Borem**, Madison Pickering, Grant Nakanishi, Maya Thumpasery, Blase Ur. "Defining Broken: UX and Remediation Tactics When Ad-Blocking or Tracking-Protection Tools Break a Website's User Experience." *USENIX Security Symposium (**USENIX Security 2023**).*